Zhehao Sun

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Education

University of British Columbia (Okanagan Campus) Kelowna, British Columbia, Canada
Master of Arts in Interdisciplinary Graduate Studies (Digital Arts and Humanities)
GPA: 91.0% (A+) May 2025 – May 2027 (expected)
Supervisor: Dr. Megan Smith *Committee:* Dr. Miles Thorogood, Dr. Patricia Lasserre

Savannah College of Art and Design (SCAD) Savannah, Georgia, USA
Bachelor of Fine Arts in Interactive Design and Game Development September 2015 – June 2018
Transferred from University of North Georgia (*Studio Art Major*, January 2013 – May 2015)

Research Interests

Game mechanics and player experience (PX); procedural content generation (PCG) and AI-assisted game design

Publications

Journal Articles

1. Xu, Y., **Sun, Z.**, Zhen, C., Lin, Y.-S., Sarker, T. H., Thorogood, M., Lasserre, P., & Dulic, A. (2026). From engagement to resilience: A systematic review of game-based learning for environmental resilience. *Sustainability*, 18(5), 2305. https://doi.org/10.3390/su18052305

Refereed Conference Proceedings & Book Chapters

1. **Sun, Z.**, Xu, Y., Zhen, C., & Smith, M. (2026). "You Were Never Meant to Win": A taxonomy of scripted loss in digital games. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY 2026)*. [Work-in-Progress] https://doi.org/10.1145/3800965.3834328

2. **Sun, Z.** (Presenter), Xu, Y., Lasserre, P., Dulic, A., & Smith, M. (2026). Re-evaluating hyper-realism in virtual reality game design through the embodied simulation gap. In *Proceedings of the 2026 Digital Games Research Association Conference (DiGRA 2026)*. Ireland. [Extended abstract] https://doi.org/10.26503/dl.v2026i4.2997

3. **Sun, Z.** (Presenter), Xu, Y., Zhen, C., Lin, Y.-S., Thorogood, M., Lasserre, P., Dulic, A., & Smith, M. (2026). Struggle as flow: Challenge, design, and experience in Soulslike games. In *Lecture Notes in Computer Science (HCI International 2026)*. Springer, Cham. https://doi.org/10.1007/978-3-032-30405-6_15

4. Zhen, C., Xu, Y., **Sun, Z.**, Dulic, A., & Thorogood, M. (2026). Sound as the score: Designing progressive audio layering as embodied music reward. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY 2026)*. [Work-in-Progress] https://doi.org/10.1145/3800965.3834330

5. Xu, Y., **Sun, Z.**, Lin, Y.-S., Lasserre, P., & Dulic, A. (2026). The absent teacher in eco-games: A theoretical framework for teacher integration in eco-games. In *Proceedings of the 2026 Digital Games Research Association Conference (DiGRA 2026)*. Ireland. [Extended abstract] https://doi.org/10.26503/dl.v2026i4.3002

6. Xu, Y., **Sun, Z.**, Zhen, C., Lin, Y.-S., Thorogood, M., Smith, M., Lasserre, P., & Dulic, A. (2026). Mapping ecological empathy: A semantic network analysis of player perceptions in 3D environmental education games. In *Lecture Notes in Computer Science (HCI International 2026)*. Springer, Cham. https://doi.org/10.1007/978-3-032-30405-6_16

Teaching & Academic Experience

University of British Columbia (Okanagan Campus) Kelowna, British Columbia, Canada
Teaching Assistant, CCS 325: AI & Creativity September 2026 – December 2026
About 200 students, 2 TAs. Lead tutorial sessions, grade, give student feedback, and facilitate in-class activities on generative AI, prompt engineering and creative coding.

Marker, CCS 330: Introduction to Game Design December 2025 – February 2026
About 120 students. Graded assignments and gave written feedback.

Teaching Assistant, ENGL 155 / DIHU 155 August 2025 – April 2026
Writing and Making with Technology in the Humanities; about 140 students, 3 TAs. Independently led a weekly mandatory discussion session, gave formative (ungraded) feedback on essays, and built the tutorial materials website.

Selected Projects

Wastescape – Research-Creation Roguelite Game (MA Thesis Project) May 2025 – Present
University of British Columbia (Okanagan Campus), Canada | Playable build on itch.io

Designed and built a playable 2D excavation roguelite game in Unity 6.5 (C#) and an early HTML5/JavaScript prototype.

Implemented deterministic procedural terrain generation, grid locomotion physics, hazard chain reactions, and an artifact archive system.

Beaver World – 3D Educational Game (Work Study) May 2026 – Present
Technical Artist / Developer | University of British Columbia (Okanagan Campus), Canada

Built a procedural vegetation generator (Unity editor tool) that fills five 4 km^2 HDRP scenes with trees, shrubs and grass from per-scene ecological rules, replacing manual placement across the game.

Fixed a severe frame-rate drop caused by over-dense terrain detail: profiled GPU passes and wrote a mesh reducer that cut grass meshes by about 87% while keeping wind-animation vertex data, reducing triangles per frame from 255 M to about 40 M.

Removed a multi-second scene-load stall and halved texture memory (3.5 GB to 1.7 GB) by baking reflection probes and removing per-object scene lookups.

Created the wildfire visuals and post-fire environment (fire spread, burnt-tree and burnt-ground materials, per-cycle regrowth); built the main menu and the shared settings system.

Destination: Imagination – VR Entertainment Ride Project January 2018 – June 2018
Environment Artist | Savannah College of Art and Design, USA

Worked in the art team on a VR amusement ride for the 2018 SCAD Day public exhibition (Featured by Unreal Engine).

Built the sci-fi cockpit of the spaceship in Unreal Engine, matched to the physical stage set made by the production design team.

Professional Experience

Art First International Education (New Channel International Education Group) China
Digital Media Instructor October 2019 – April 2025

Taught Unreal Engine development, computer graphics fundamentals, and game design principles; guided portfolio and demo reel development for students applying to international universities.

CATS Co., Ltd. (NTT Data) Shinyokohama, Japan
3D Designer September 2018 – September 2019

Contributed to an autonomous vehicle technology visualization project through cross-functional collaboration with programming and technical teams.

Developed real-time 3D data visualization systems using Unreal Engine for automotive sensor data representation.

Implemented web-based 3D rendering solutions utilizing Java, WebGL, and C# technologies.

Xi'an Kushi Information Technology Co., Ltd. Xi'an, China
3D Artist Intern June 2018 – September 2018

Participated in augmented reality application development using Unity and Vuforia platforms.

Created 3D models and textures for virtual environment assets.

Technical Skills

Game Engines: Unreal Engine (Blueprints, custom shaders, real-time VFX), Unity (C#, UI Toolkit, 2D/3D/VR), Vuforia
Programming & Web: C#, JavaScript / HTML5, Three.js / WebGL, Java
3D Production: Modeling, sculpting, retopology, UV mapping, texturing, rigging, animation, environment lighting
Research & Design: Game design, player experience, computer graphics, UX for interactive media

Honors & Awards

UBC Okanagan Graduate Research Scholarship (OGRS) ($5,000/year; $10,000 total)	2025 – 2027
SCAD Dean's List	2017 – 2018
SCAD Academic Scholarship ($10,000/year)	2015 – 2018
SCAD Student Incentive Scholarship	2015 – 2018